

EXP 2: Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

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hemmm case 33.sci (C:\Users\DELL\Documents\hemmm case 33.sci) - SciNotes
exp 6 emf.sci  exp 3.sci  exp 4.sci  exp 5.sci  hem exp 6.sci  exp 6.sci
vishnu exp 5.sci  hem exp 07.sci  hem exp 8.sci  vis 6.sci  vis exp 7.sci  vis 8.sci  hemmm case 2.sci  *hemmm case 33.sci

1 clear;
2 clear;
3 q=input('Enter the value of q:');
4 r=input('Enter the value of r:');
5 epsilon=8.854*10^-12;
6 pi=3.14;
7 E=q/(4*pi*epsilon*r^2);
8 disp('The electric field is:', [E]);
9 D=E*epsilon;
10 disp('The flux density is:', [D]);
11 x = linspace(0, 10, 50);
12 y = E * exp(-0.1 * x);
13 z = D * exp(-0.1 * x);
14 figure();
15 subplot(2, 2, 1);
16 plot2d3(x,y);
17 xlabel('R');
18 ylabel('Electric field intensity');
19 title('Electric field intensity');
20 subplot(2, 2, 2);
21 plot2d3(x,z);
22 xlabel('R');
23 ylabel('Electric flux intensity');
24 title('Electric flux intensity');

```

Line 24, Column 41.

SciLab 2024.1.0 Console

File Edit Control Applications ?

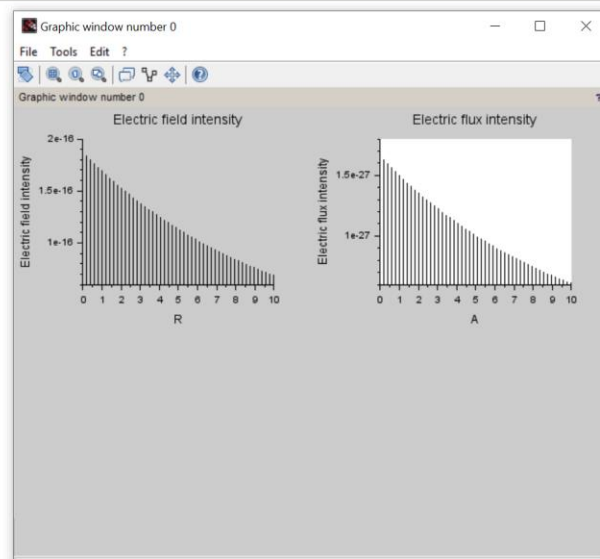
SciLab 2024.1.0 Console

```

Enter the value of q=3*10^-6
Enter the value of r=3

"The electric field is"
1.877D-16
"The flux density is"
1.662D-27
exec: Wrong number of output argument(s): 0 expected.
-->

```



Ex 101-2

Determination and plotting of electric field
flux density is charges with varying radius

$$Q = 3 \times 10^{-6}$$

$$r = 3 \text{ cm}$$

$$\text{Electric field } E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$\text{flux density } D = E \cdot \epsilon_0$$

The electric field is

$$1.8770 \times 10^{-16}$$

The flux density is

$$1.6620 \times 10^{-27}$$

Electric field

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

$$E = \frac{3 \times 10^{-6}}{4\pi(8.854 \times 10^{-12})(3 \times 10^{-2})^2}$$

$$E = \frac{3 \times 10^{-6}}{4\pi(8.854 \times 10^{-12})(9 \times 10^{-4})}$$

$$E = 1.8770 \times 10^{-9}$$

flux density

$$D = \epsilon_0 E$$

$$D = (8.854 \times 10^{-12})(3.00 \times 10^{-9})$$

$$D = 1.6620 \times 10^{-20}$$

$$D = 1.6620$$

